Reg. No.															

B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024
Fourth Semester
21CSE253T - INTERNET OF THINGS
(For the candidates admitted from the academic year2021-2022 to 2023-2024)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours | Max. Marks: 75

PART - A (20 x 1 = 20 Marks)
Answer **ALL** Questions

	Question	Marks	BL	CO	PO
1.	The LoRaWAN protocol is developed and maintained by the A) ISO/OSI B) Telecommunication department C) LoRa Alliance D) ARPANET	1	1	1	6
2.	In which type of topology, the LoRaWAN networks are deployed? A) Ring topology B) Star topology C) Star-of-Stars topology D) Hybrid topology	1	1	1	6
3.	The piece of software running on a server that processes join-request messages sent by end devices. What is the name of that server? A) Network server B) Application server C) Gateway server D) Join server	1	1	1	1
4.	The indoor gateways of LoRaWAN are also referred as, A) macrocell B) picocell C) cluster head D) access point	1	1	1	1
5.	What modulation scheme is followed by the standard IEEE 802.15.4g? A) OFDM B) CSMA/CA C) CSMA/CD D) TDMA	1	1	2	2
6.	Which standard falls under the IEEE 802.11 family of wireless communication standards, specifically designed for Low Power, Wide Area (LPWA) networks, and it is often associated with IoT (Internet of Things) applications? A) IEEE 802.10ah B) IEEE 802.11ah C) IEEE 802.11e D) IEEE 802.10e	1	1	2	2
7.	The protocol that is adapted for constrained networks, providing lightweight, publish-subscribe communication is the, A) MQTT protocol B) LoRaWAN protoco C) CoAP protocol D) IPv4 protocol	1	1	2	2
8.	IoT devices with resource limitations, such as low processing power, limited memory, energy constraints, and often restricted communication capabilities are called as A) Constrained networks B) Constrained nodes C) Relay nodes D) Gateway nodes	1	1	2	2

9. Select the IoT model specification that defines the physical entities, virtual entities, devices, resources and services in the system. 1 1 3 1

A) Domain model specification B) Process model specification
C) Service mspecification D) Functional view model specification

10. Pick the digital sensor (only) category form the given options. 1 1 3 1

A) Piezoelectric sensor B) Color sensor
C) Smoke sensor D) Alcohol sensor

11. What is the operating input voltage of Arduino board? 1 1 3 1

A) 3.3 V B) 13 V
C) 10 V D) 12 V

12. Which version of RaspberryPi includes GPIO connector and the header pins are not soldered? 1 1 3 1

A) Raspberry Pi 1 B) Raspberry Pi 2
C) Pi Zero D) Raspberry Pi 3

13. A powerful network management protocol that provides a standardized way to configure, manage, and monitor network devices is, 1 1 4 2

A) NETCONF B) YANG
C) Django D) Xively

14. In order for the businesses to accelerate their IoT initiatives, reduce development and operational costs, and unlock new revenue streams through innovative connected products and services, you need to leverage ________________ . 1 1 4 2

A) Xively cloud B) Power supply
C) Resources D) Data Center

15. Pick up the feature not supported by Edge Streaming Analysis and Network Analytics. 1 1 4 2

A) Real-Time Monitoring B) Distributed Intelligence
C) Insight Generation D) Central Intelligence

16. Which among the following is the lightning-fast, open-source, and unified analytics engine for large-scale data processing? 1 1 4 2

A) Apache Kafka B) Apache Spark
C) Hadoop ecosystem D) R programming

17. Which platform supports MQTT and REST protocols for applications, devices, gateways, and administrative tasks?. 1 1 5 4

A) Python platform B) R programming platform
C) Gateway platform D) IBM Watson IoT platform

18. The next phase of collaboration which represents continuing IACS-to business convergence is, 1 1 5 4

A) CpWE B) CoWE
C) VoIP D) IaaS

19. The ______________ approach allows cisco services to help build the architecture on a project-by-project basis as utilities implement smart grid. 1 1 5 2

A) MQTT B) modular Gridblock
C) CpWE D) CoWE

20. In which architecture layer, the different services are defined and provided to different users? 1 1 5 2

A) Application layer B) Middleware layer
C) Networking layer D) Sensing layer

PART - B (5 x 8 = 40 Marks)
Answer **ALL** Questions

	Question	Marks	BL	CO	PO
21 a.	Explain about the Fog, Edge and Cloud in IoT.	8	2	1	6
	(OR)				
b.	Explain in detail about the functional blocks of IoT ecosystem.	8	2	1	1
22 a.	Describe the protocol IEEE 802.11ah with appropriate specifications.	8	2	2	2
	(OR)				
b.	Explain in detail about the application transport methods.	8	2	2	2
23 a.	Draw the architecture of Raspberry Pi and explain the utilization of all the pins in details. Also compare its features with Arduino board.	8	2	3	1
	(OR)				
b.	Describe about the IoT system building blocks in detail.	8	2	3	1
24 a.	Discuss in detail about the Xively cloud for IoT.	8	2	4	2
	(OR)				
b.	Discuss the need for Django in IoT environment.	8	2	4	2
25 a.	With appropriate architecture explain the Gridblocks reference model.	8	2	5	4
	(OR)				
b.	Define CPwE. Explain its significance in IoT environment.	8	2	5	4

PART - C (1 x 15 = 15 Marks)
Answer **ANY ONE** Question

	Question	Marks	BL	CO	PO
26.	With necessary diagrams explain the IoT architectures in detail.	15	2	1	1
27.	Explain in detail about the application layer protocols CoAP and MQTT. Also compare their significance.	15	2	2	2

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